General Chemistry 1
Problem Set#5
Gas Laws

Name:
Strand and Section:

Instructions: Write the detailed solutions of every problem on your activity notebook.

1. A sample of CO2 gas at 22 ˚C and 1.00 atm has a volume of 2.00 L. Determine the ratio of the original volume to the final volume when
   a. The pressure and amount of gas remain unchanged and the Celsius temperature is doubled.
   b. The pressure and amount of gas remain unchanged and the Kelvin temperature is doubled.
2. A tank is filled with a gas to a pressure of 977 mm Hg at 25 ˚C. When the tank is heated, the pressure increases to 1.50 atm. To what temperature was the gas heated?
3. A cylinder with a movable piston records a volume of 12.6 L when 3.0 mol of oxygen is added. The gas in the cylinder has a pressure of 5.83 atm. The cylinder develops a leak and the volume of the gas is now recorded to be 12.1 L at the same pressure. How many moles of oxygen are lost?
4. A piece of dry ice (CO2) has a mass of 22.50 g. It is dropped into an evacuated 2.50-L flask. What is the pressure in the flask at -4 ˚C.